Universitat Politècnica de Catalunya
Facultat d'Informàtica de Barcelona (FIB)

End-to-End Optical Music Recognition for Full Music Sheets using Deep Learning

GEP Deliverable 1: Context and Scope

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Specialty: Computació

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1 Introduction and Context

1.1 Academic/Professional Framework

This project is developed in the academic context of the Bachelor's Thesis (TFG) at FIB–UPC, under modality A, as part of the requirements to complete the Computer Science degree at the Universitat Politècnica de Catalunya (UPC). The project is supervised by Manel Frigola Bourlon, professor at the Facultat d'Informàtica de Barcelona (FIB), and guided by Xavier Morales Sorolla as GEP tutor.

The work is framed within the Computació specialty. The topic was selected due to my interest in the intersection of music and technology, and because of the potential impact of Optical Music Recognition (OMR) in areas such as music education, digital archiving, and music information retrieval.

1.2 Domain Concepts and Terminology

Optical Music Recognition (OMR): The process of automatically converting scanned or photographed music sheets into machine-readable symbolic formats (e.g., MusicXML, MIDI).

Music Sheet/Score: A printed or digital document containing musical notation, including notes, staves, time signatures, and other musical symbols.

End-to-End System: A unified approach that maps input directly to output without intermediate manual steps, typically using a single neural network or an integrated pipeline.

Music Information Retrieval (MIR): The interdisciplinary field focused on extracting meaningful information from music data.

Symbolic Notation: A machine-readable representation of music (e.g., MusicXML, MIDI), in contrast to audio waveforms.

Digital Archiving: The process of converting and preserving documents in digital form for long-term accessibility.

In addition to these core concepts, the project relies on basic music theory (notation, rhythm, and structure) and technical foundations in deep learning, computer vision, and data processing. When required, these notions will be introduced and explained to keep the document accessible to readers with different backgrounds.

1.3 Problem Statement

As a musician, I have repeatedly faced the need to digitize paper scores for sharing, archival purposes, and use in composition or analysis software. This is especially relevant in rehearsal and live-performance contexts, where musicians may receive a physical chart and need to perform it immediately.

In these situations, paper limits flexibility. Performers may need to transpose parts for different instruments, generate quick audio playback to understand style and groove, or store the score in a searchable digital library (e.g., on a tablet). Manual transcription is slow and error-prone, which motivates the need for an efficient and accurate method to convert physical music sheets into machine-readable formats.

1.4 Stakeholders

- **Musicians:** Readers and performers who benefit from digital scores for practice, performance, sharing, and transposition.

- **Music Educators:** Teachers who use digital sheet music in class and provide interactive learning resources to students.
- **Music Archivists:** Professionals responsible for preserving and organizing collections, who benefit from improved accessibility and long-term preservation.
- **Music Software Developers:** Companies and independent developers who can integrate OMR outputs into notation, playback, or learning applications.
- **MIR Researchers:** Researchers in music information retrieval who can use OMR data for analysis, recommendation systems, and musicology studies.
- **General Public:** Music enthusiasts interested in accessing and interacting with digital music sheets for personal use and learning.

2 Justification of the Chosen Alternative

2.1 State of the Art / Existing Alternatives

Current OMR literature distinguishes three broad families of approaches:

1. **Traditional modular pipelines (e.g., rule-based systems):** They decompose the task into fixed stages (pre-processing, staff detection/removal, symbol segmentation, symbol classification, and reconstruction). Their main weakness is error propagation between stages.
2. **CRNN-CTC approaches:** They combine convolutional features with recurrent sequence modeling and CTC training, adapting methods from text recognition. They are effective for simple notation lines but can struggle when dense polyphony and complex page structure are present.
3. **End-to-end transformer approaches:** They formulate OMR as image-to-sequence translation with attention-based encoder-decoder architectures. This family is currently the most promising for full-page and structurally complex scores.

2.2 Comparison Criteria

The alternatives are compared using criteria aligned with the scope of this TFG:

- **Accuracy and robustness:** Resistance to image noise, quality variation, and notation complexity.
- **Polyphony support:** Ability to represent simultaneous notes and multiple voices correctly.
- **Layout awareness:** Capacity to process complete pages with multiple systems and non-trivial reading order.
- **Semantic completeness:** Generation of outputs that preserve musical meaning (pitch, duration, meter, and key context).
- **Implementation feasibility:** Compatibility with available time, compute resources, and development effort for a Bachelor's thesis.
- **Evaluation compatibility:** Possibility of benchmarking with public datasets and reproducible metrics.

2.3 Comparative Analysis

Traditional pipelines provide interpretability and low entry cost, but their stage-by-stage structure tends to amplify upstream errors and requires substantial manual tuning for new score conditions.

CRNN-CTC models simplify training and can work well on constrained settings (e.g., single staves or reduced vocabulary). However, they are less suited to rich polyphony and full-page structure when direct sequence alignment becomes ambiguous.

Transformer-based models offer the strongest overall balance for this project. Their attention mechanisms support broader visual context and improve handling of long-range dependencies (for example, symbols whose meaning depends on earlier context). In addition, they integrate naturally with compact tokenized output formats suitable for sequence generation.

2.4 Selected Alternative and Rationale

The selected alternative is an **end-to-end transformer architecture**, following the **Sheet Music Transformer (SMT)-style** formulation with **Humdrum** `**kern` tokenization.

Rationale:

1. **Better fit for full-page transcription:** Attention-based decoding is more appropriate for non-local dependencies and dense notation than strictly monotonic sequence assumptions.

2. **Good representational efficiency:** The `**kern` format is compact and expressive enough to encode relevant symbolic information for this stage of the project.

3. **Feasible implementation path:** A medium-scale encoder-decoder model is realistic within TFG constraints, both in training cost and engineering effort.

4. **Strong baseline potential:** The approach can be evaluated with common OMR metrics and compared against prior deep-learning baselines.

2.5 Constraints and Decision Boundaries

- **Data availability:** High-quality ground truth for full pages is limited; therefore, synthetic data and careful validation splits are required.

- **Sequence length:** Dense pages produce long output sequences, so compact encodings (e.g., `**kern`) are preferred over verbose alternatives.

- **Computational resources:** Training must fit available GPU memory and time constraints, which limits model size and batch configuration.

- **Project scope:** The objective is a solid and reproducible prototype, not a production-ready universal OMR engine.

- **Too ambitious** approaches (e.g., multi-modal fusion, large-scale pretraining) could be out of scope for this TFG stage but could be future extensions, then a CRNN-CTC for monophonic sheets or a modular pipeline for simple scores could be fallback alternatives if the transformer approach proves infeasible within the project timeline.

3 Project Scope

3.1 General Objective

State the main objective of the TFG clearly and concisely.

3.2 Specific Objectives

List measurable sub-objectives that decompose the general objective.

3.3 Functional Requirements

Describe what the solution must do (features, behaviours, outputs).

3.4 Non-Functional Requirements

Describe quality attributes and constraints (e.g., accuracy, speed, usability, robustness, portability).

3.5 Scope Boundaries and Exclusions

Specify what is in scope and explicitly what is out of scope for this deliverable/TFG stage.

3.6 Risks and Potential Obstacles

Identify potential risks and obstacles (technical, data-related, planning, integration) and their expected impact.

4 Methodology and Rigour

4.1 Project Methodology

Describe the chosen methodology (e.g., classical/cascade, agile, hybrid) and justify why it fits the project.

4.2 Adaptation to This Project

Explain how the methodology is tailored to your concrete TFG context and constraints.

4.3 Execution Approach

Describe the intended phases/iterations at a high level and the expected deliverables/results of each one.

4.4 Monitoring and Follow-up Tools

Describe how progress will be monitored (meetings, milestones, indicators, planning board, repositories and logs).

5 Document Organisation

Briefly explain the structure of this document and what each section contains.

6 Conclusions

Summarize the context, chosen alternative, defined scope, and methodological approach. Highlight the expected contribution of the TFG.